

BUMK 744

REAL ESTATE MARKET SEGMENTATION ANALYSIS

PCA & K-MEANS CLUSTERING TO IDENTIFY HOUSING
MARKET SEGMENTS

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Introduction

Business Context

Purpose of Analysis



Real estate markets are highly heterogeneous



Traditional segmentation (price, bedrooms, etc.) can be simplistic



Investors and agents need data-driven market structure insights

Objective

move beyond price prediction → uncover natural housing market segments



DATASET OVERVIEW

Data Source

- Residential Property Transactions (Kaggle)
- 21,613 property transactions (May 2014-May 2015)
- 21 features per listing
- Rich Feature Diversity: latitude, longitude, geography

Research Objectives

- Reduce high-dimensional housing data using PCA
- Identify natural housing segments via K-Means
- Uncover what property features define each segment



KEY VARIABLE SUMMARY

01

TARGET VARIABLE

Price

- Range: \$75K – \$7.7M
- Median: \$450K
- Mean: \$540K (right-skewed distribution)

02

SIZE METRICS

- sqft_living (290 – 13,540 sq ft)
- sqft_lot, sqft_above, sqft_basement

03

PROPERTY CHARACTERISTICS

- Bedrooms (0–33), bathrooms (0–8), floors
- Year built (1900–2015)

QUALITY AND LOCATION FEATURES

01

QUALITY INDICATORS

- Grade (construction quality)
- Condition (maintenance level)
- Ordinal scoring system

02

LOCATION VARIABLES

- Latitude / Longitude
- 70 ZIP codes
- Waterfront + view scores
- Only ~0.8% waterfront properties (rare premium segment)

CORE GOAL

Identify meaningful housing market segments that are not visible through simple price grouping



ANALYTICAL PURPOSE

WHY NOT TRADITIONAL METHODS

- Price bands oversimplify heterogeneity
- High correlation between features distorts interpretation

SOLUTION

- PCA → structure reduction
- K-Means → segmentation discovery





MANAGERIAL VALUES

PCA

WHY?

- Simplifies complex housing feature space
- Removes redundancy between variables
- Enhances clustering stability

PCA

STRATEGIC APPLICATIONS

- More reliable segmentation inputs
- Easier interpretation of underlying housing dimensions (size, quality, location)

CLUSTERING

WHY?

- Identifies real estate “market types” rather than individual listings

CLUSTERING

STRATEGIC APPLICATIONS

- Pricing strategy by segment
- Targeted marketing for buyers/investors
- Property recommendation systems
- Investment screening for developers

ANALYSIS OVERVIEW



PC1

Size / Wealth

Captures sqft_living,
price, bedrooms,
bathrooms — the scale
of the property

PC2

Location / Luxury

Captures waterfront,
view, floors — the
premium & location
value

RESULT

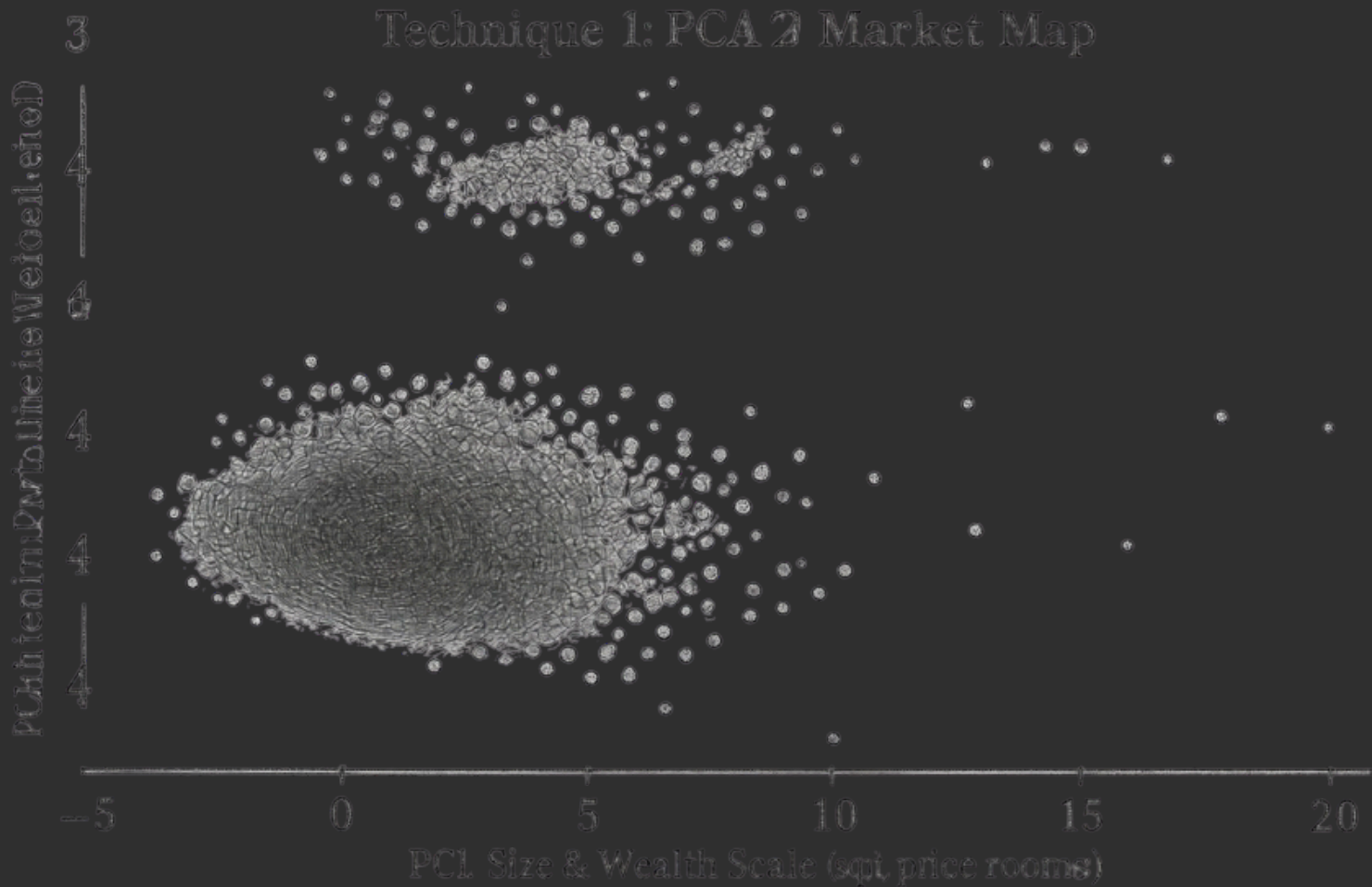
2D Market Map

Entire housing dataset
plotted in 2
dimensions, ready for
clustering

TECHNIQUE 1: PCA

WHAT PCA DOES:

- Reduces 9 features → 2 principal components
- Handles multicollinearity (sqft variable overlap)
- Converts features into uncorrelated components



TECHNIQUE 2: K-MEANS CLUSTERING

ELBOW METHOD

Tested $K = 1$ to 20 → Optimal $K = 4$

- Groups properties into natural market segments
- Based on PCA-transformed components
- Identifies clusters such as:
 - Entry-level housing
 - Suburban family homes
 - Luxury waterfront estates



ENTRY-LEVEL

Small homes, lower price, budget-conscious buyers



MID-RANGE

Average size & price — core market segment



UPPER-MID

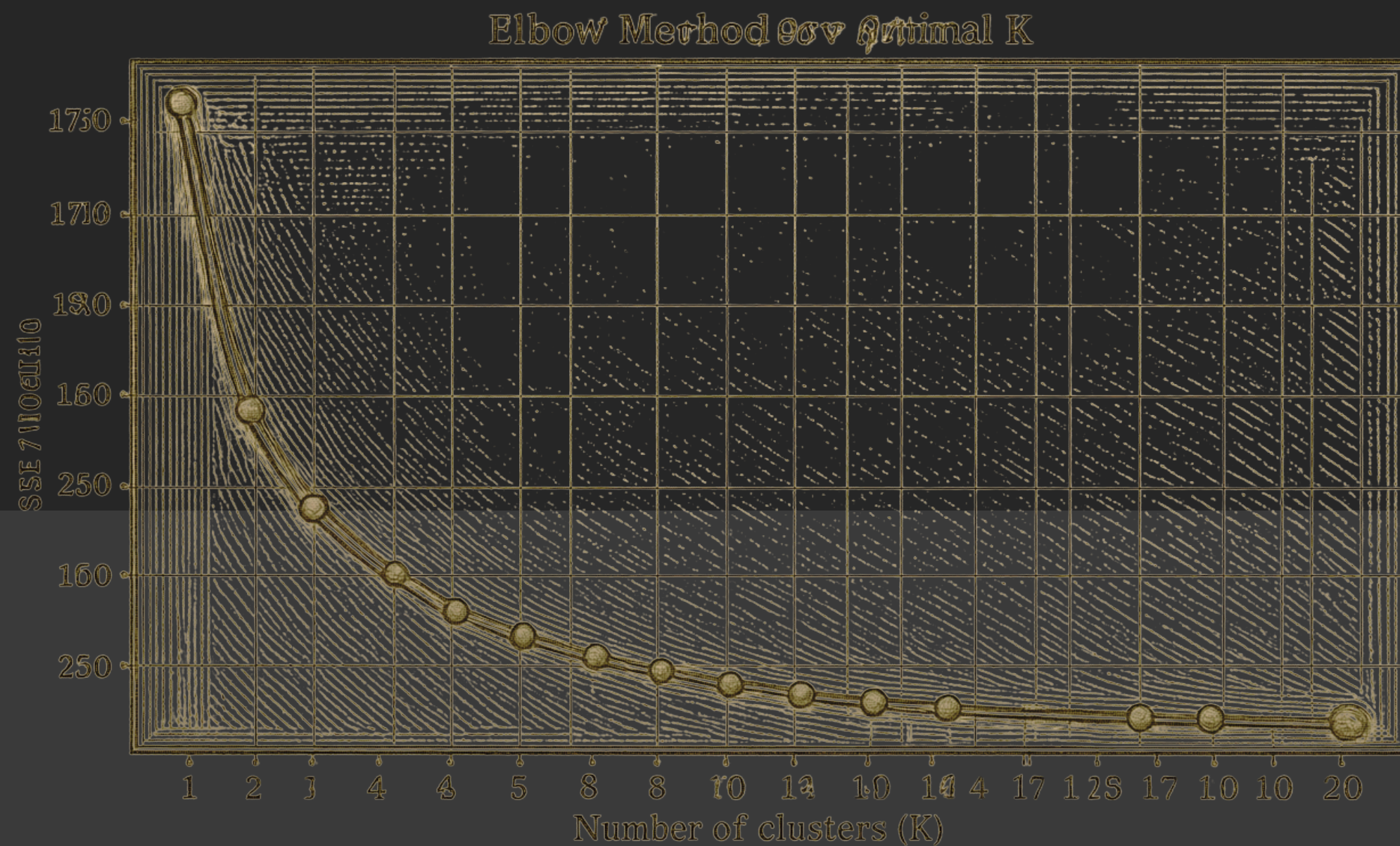
Larger sqft, better location scores



LUXURY

High PC2 — waterfront, views, premium finishes

ELBOW METHOD FOR OPTIMAL K



- Elbow: $k=4$
- Clear elbow point
- Strong cluster separation
- High business interpretability

K = 4 provides the most interpretable and actionable segments



CLUSTER PROFILES

WHAT EACH CLUSTER REPRESENTS:

01
ENTRY-LEVEL

PC1: Low (-2 to 0)
PC2: Low (0 to 1)

High-volume, affordable homes. Smallest sqft, fewest bedrooms. Primary target for first-time buyer campaigns.

02
MID-RANGE

PC1: Mid (0 to 2)
PC2: Low-Mid (0 to 1.5)

Core of the market. Standard layout, consistent pricing. Broadest buyer segment — ideal for general marketing.

03
UPPER HIGH
DENSITY PROPERTIES

PC1: High (2 to 6)
PC2: Mid (0 to 3)

Larger homes with better location scores. Buyers seeking space and amenities. Strong upsell opportunity.

04
LUXURY

PC1: Variable
PC2: High (3 to 10)

Low volume, high value. Waterfront and view properties. Premium positioning required — differentiated marketing.

Key Insight: Market is structured by combinations of size, quality, and geography—not price alone



STRATEGIC IMPLICATIONS

1

REAL ESTATE AGENTS

- Better property positioning and pricing strategy

2

INVESTORS

- Identify undervalued or high-growth segments

3

DEVELOPERS

- Spot underserved housing categories

4

POLICY-MAKERS

- Identify aging or low-quality housing zones for redevelopment

CONCLUSION & RECOMMENDATIONS

Key Insights:

PCA → Size/Wealth &
Location/Luxury Dimensions

Note: Preliminary

Results for Cluster Summaries

1

SEGMENT-SPECIFIC MARKETING

1

- Tailor campaigns by cluster — budget messaging for Cluster 0, luxury positioning for Cluster 3

2

UPSELL CLUSTER 2

2

- Prioritize Upper-Mid listings for upsell opportunities — highest growth potential segment

3

GEOGRAPHIC TARGETING

3

- Use zip codes and lat/long cluster data to guide regional investment and inventory allocation

01 SCOPE ACHIEVED

- Moves beyond descriptive pricing
- Creates actionable market segmentation
- Supports data-driven real estate decision-making

KEY TAKEAWAYS

Analysis reveals the hidden structure of the housing market

02 SCOPE ACHIEVED

- Understanding how the market is structured is more valuable than simply predicting prices.



THANK
YOU

QUESTIONS